CHARBEL DIB

AI / Machine Learning Engineer | NLP, RAG & AI agents

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PROFILE

ENSTA engineering graduate specialising in NLP, RAG and model evaluation. Two applied AI internships and projects spanning models, APIs and interfaces. First author of an IEEE paper. Seeking a permanent AI / machine learning role in France; available immediately.

EDUCATION

ENSTA, Brest campus | Engineering degree (Master's level) | 2024 - 2026

Specialisation: Observation Systems and Artificial Intelligence.

Coursework: machine learning, deep learning, computer vision, signal processing and data fusion.

EXPERIENCE

Astek Innovation Lab | Final-year internship - NLP & RAG

La Défense, France · Mar - Sep 2026

- Designed an evaluation framework for climate fact-checking across retrieval, evidence extraction and classification; analysed error propagation between stages.
- Compared chunking strategies, hybrid DPR + BM25 retrieval and rerankers; evaluated extraction prompts and a Phi-3 classifier previously adapted with LoRA.
- Organised experiments using shared metrics (NDCG, recall, macro-F1), Docker, caching and logs to support reproducibility and comparison.

CESI / LINEACT | AI & machine learning internship

Nanterre, France · May - Aug 2025

- Built an experimental dataset and compared predictive models and clustering to study relative density in L-PBF additive manufacturing; work led to a first-author IEEE paper.

SELECTED PROJECTS

FleetGuard | XGBoost, scikit-learn, FastAPI, React | 2026

- Built a Scania APS failure-classification pipeline with calibration and a cost-based decision threshold. Official test set of 16,000 cases: 95.73% recall and 43.05% precision.
- Developed an API and a batch CSV review interface; documented results in a model card.

ServicePilot | LangGraph, FastAPI, SQLite, FAISS | 2026

- Created a support SDK and prototype with human approval before ticket creation, server-side permission checks and resumable workflows; added tests and CI.

ENSTA Assistant | RAG, multilingual-E5, FAISS | 2026

- Built a bilingual prototype over web/PDF/XLSX sources with citations, refusal when evidence is insufficient, a FastAPI service and daily web-index updates.

SKILLS

ML / NLP: Python, PyTorch, scikit-learn, XGBoost, Transformers, embeddings, RAG, reranking.
Data / software: Pandas, NumPy, SQL, FastAPI, LangGraph, FAISS, Docker, Git, Linux.

PUBLICATION & AWARD

First author: *Cluster-Based Learning for Relative Density Prediction in Laser Powder Bed Fusion.* IEEE 3SCEA, 2026. DOI: 10.1109/3SCEA68071.2026.11603067

1st prize, SPASEA 2025: posAIdon, a team-built maritime copilot; Naval Group / DGA / ENSTA.

Languages: French fluent · English fluent · Arabic native.